(45 miles per hour). If the enable 1 circuit is deactivated after the rear or the front and rear axle lockers are engaged, all outputs from the axle lock switch are dropped causing both axles to unlock. The enable 2 circuit is automatically deactivated whenever the vehicle speed is greater than about 16 kilometers per hour (10 miles per hour). If the enable 2 circuit is deactivated after the rear or the front and rear axle lockers are engaged, the outputs from the axle lock switch are unaffected and the locked axles remain locked. However, an unlocked axle cannot be locked until the vehicle speed is reduced and the enable 2 circuit is again activated.

In addition, once activated, both enable circuits will remain active regardless of the status of the ignition switch input. Therefore, any locked axle will remain locked and the various components of the axle locker system will remain functional after the ignition switch is turned to the Off position. However, while the currently selected axle locker mode remains active with the ignition switch turned Off, if the key is removed from the ignition lock cylinder, Off is the only other axle locker mode that can be selected with the axle lock switch. For as long as the key is removed from the ignition lock cylinder, the cluster logic will interpret any revision to the input status of either request circuit from the axle lock switch as a cancellation request and will deactivate the enable 1 circuit and all outputs from the axle lock switch are dropped, causing both axles to unlock. Otherwise, once locked, any locked axle will remain locked until the axle lock switch is deactivated (enable 1 circuit is deactivated), or until the Off mode is manually selected by moving the axle lock switch rocker to the Off position. The EMIC also provides the vehicle operator with distinct visual and/or audible indications as to the current status of the axle locker system through chime warnings and illumination of the rear/front lock indicators in the cluster as outlined elsewhere in this service information.

The hard wired input and output circuits of the EMIC axle locker control may be diagnosed using conventional diagnostic methods. However, proper testing of the EMIC programming and the electronic vehicle speed and transfer case shift position switch messages received by the EMIC over the PCI data bus requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information.

CHIME SERVICE

The EMIC is equipped with hardware and software to provide chime service for all available features in the chime warning system. Upon receiving the proper chime inputs, the EMIC activates an integral on-board audible tone generator to provide audible chime tones to the vehicle operator. The chime tone generator in the EMIC is capable of producing single chime tones or repeated chime tones at two different rates: a slow rate of about fifty chime tones per minute, and a fast rate of about 180 chime tones per minute. The internal programming of the EMIC determines the priority of each chime tone request input that is received, as well as the rate and duration of each chime tone that is to be generated.

The EMIC relies upon hard wired inputs from the door ajar switches, the left multi-function switch, the ignition switch, and the park brake/brake warning indicator switches to provide chime service for the driver/passenger door ajar warning, the head/park lights-on reminder, and the key-in ignition reminder. For the remaining chime warning functions, the EMIC uses a combination of hard wired inputs, electronic message inputs received from other modules over the PCI data bus, and internal programming. (Refer to 8 - ELECTRICAL/CHIME/BUZZER - OPERATION).

The hard wired chime inputs to the EMIC may be diagnosed using conventional diagnostic methods. However, proper testing of the EMIC programming and the electronic chime request messages received by the EMIC over the PCI data bus requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information.

ELECTRONIC PINION FACTOR

Each time the EMIC receives an electronic pinion factor request message from the Powertrain Control Module (PCM) over the PCI data bus, it provides a response to the PCM containing the appropriate electronic pinion factor information. Because there are numerous optional combinations of tire size, axle ratio, and transfer case type which are available for this vehicle, the electronic pinion factor needs to be considered for accurate processing of the vehicle speed and distance information. The instrument cluster stores the electronic pinion factor information and broadcasts it over the PCI data bus upon request by the PCM. Using this factor and the Vehicle Speed Sensor (VSS) input, the PCM then calculates the proper vehicle speed and distance information, and transmits the appropriate electronic vehicle speed messages over the PCI data bus for use by other electronic modules in the vehicle.

Proper testing of the PCI data bus and the electronic data bus message inputs to and outputs from the EMIC that control the electronic pinion factor function requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information.

INTERIOR LIGHTING CONTROL

The EMIC contains an integral timer and logic circuit to perform both timer and control functions for the interior courtesy lamps. The EMIC uses hard wired inputs from the ignition switch, both door ajar switches on separate driver and passenger door ajar